



A Unified Multimodal System for Human Motion Capture

Arun Tamura

Advisor: Prof. Amin Arbabian · Mentor: Shubo Yang · Arbabian Lab, Stanford University

1 · Introduction

Why motion, and what we're building

- Camera-based systems capture rich spatial detail, but fail under **occlusion**, **poor lighting**, or when **someone steps out of frame**—exactly when healthcare monitoring needs them most.
- Radar-based motion sensing **stays robust through all three failure modes**—the basis for continuous, unobtrusive monitoring instead of a periodic clinic visit.
- We're building a large-scale gait dataset (walking, freezing, shuffling, sit-to-stand) with **SMPL ground truth** from four **synchronized cameras**, not hand-labeled annotations.
- Goal: predict gait activity from radar motion signatures**—passive, non-invasive monitoring in real homes and clinics, not just in a lab.

Prior work: our lab's MVDoppler-Pose (CVPR 2025) showed radar-only gait classification under controlled, indoor conditions via multi-view mmWave sensing.

My prior quarter: I onboarded to the lab's radar system and redesigned its indoor waveform, nearly **doubling sweep bandwidth (2→4 GHz)** and **duty cycle (60%→99%)**.

The tradeoff above, quantified:

Sensor	Spatial Detail	Motion/Velocity	Robustness to Occlusions	Robustness to Lighting	Range
Camera	High	×	×	×	Medium
mmWave Radar	Medium	✓	✓	✓	High
UWB Radar	Low	✓	✓	✓	Short

Table 1. Cameras win on spatial detail; radar wins on motion sensitivity and robustness to occlusion/lighting — the basis for our sensor choice.



Fig. 1. Healthcare monitoring: exactly where occlusion and lighting break camera-only systems.



Fig. 2. Robotics & human-aware interaction — motion sensing enables safe, adaptive response to people.

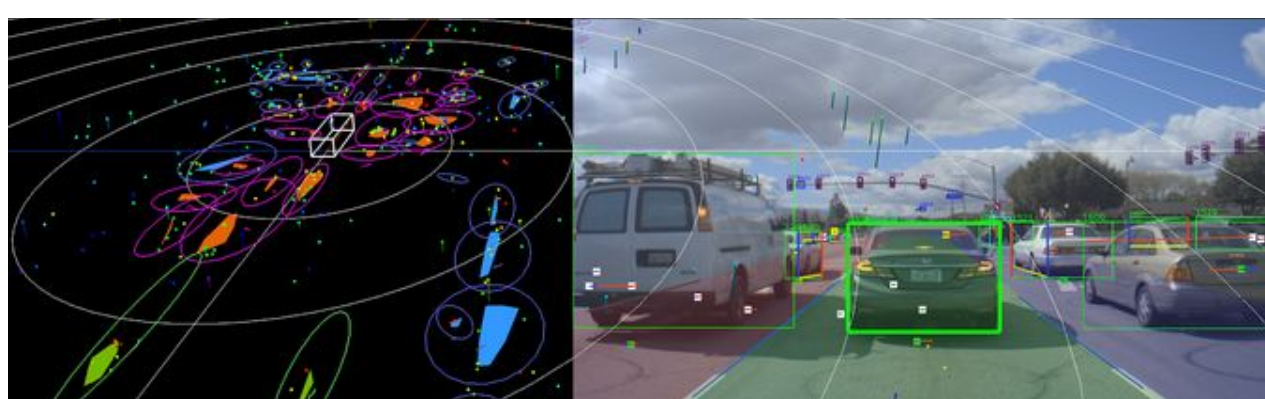


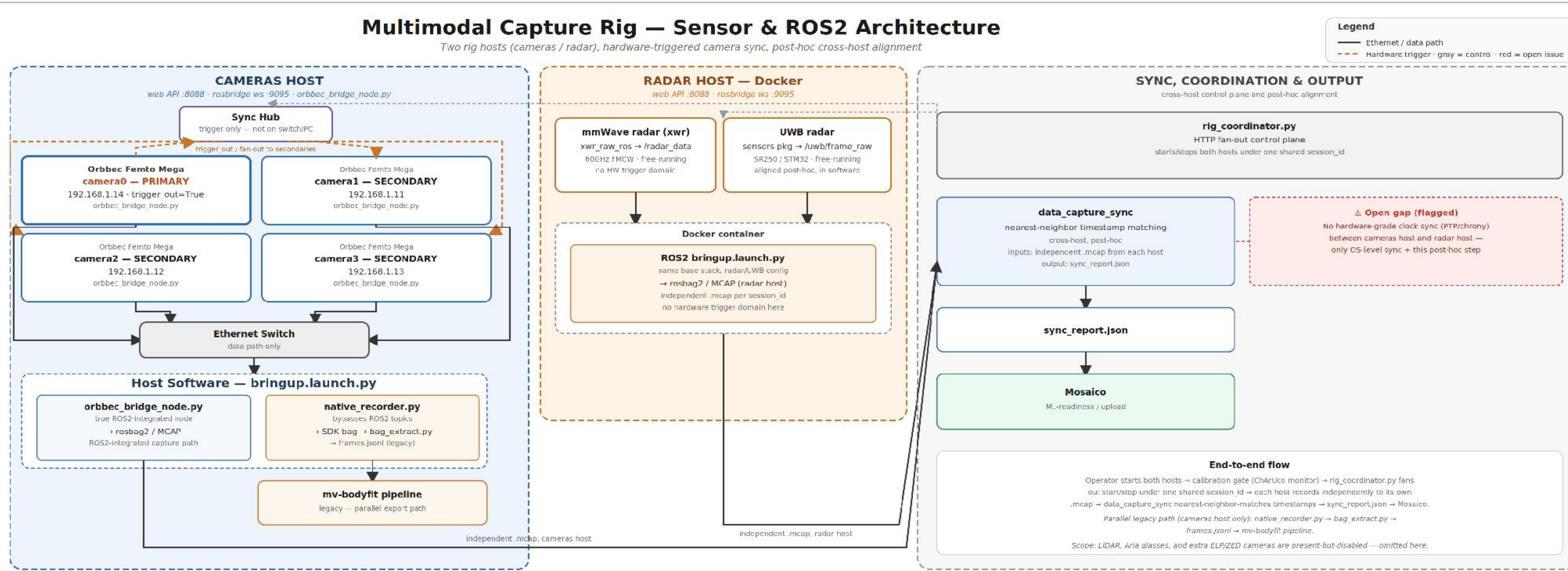
Fig. 3. Radar-based motion sensing is already proven at scale in autonomous driving — the same occlusion/lighting robustness this project extends to healthcare.



2 · Methods: Integrated System & Ground-Truth Pipeline

The integrated capture system

- The system unifies four Orbbec Femto Mega cameras, a TI IWR6843ISK mmWave radar, and a Bosch UWB radar **into one ROS2 system**—replacing an earlier two-computer setup that needed manual timestamp alignment.
- A shared `/capture/enable` trigger opens and closes every sensor's recording window at the same instant, landing all six devices in **one combined, time-aligned rosbag per session**.
- The four cameras additionally sync to each other via a **hardware trigger** through Orbbec's sync port; the radars and the cross-host handoff rely on the **software `/capture/enable` signal** alone.
- Piloted on a small group; now scaling to 20 subjects for a publication-ready dataset.



Ground-truth fitting pipeline

- We fit SMPL directly to **raw 2D keypoints from all 4 camera views at once**, not to a single pre-triangulated 3D point (our original approach).
- Problem: triangulating first collapses 4 views into one 3D point before fitting starts—a wrong point (from occlusion or fast motion) was undetectable, causing a recurring **spin/contortion bug during fast turns**.
- Fix: fitting to all 4 raw views lets the optimizer see when cameras disagree, eliminating the bug. Validated against an independent shoulder-based heading estimate, **correlation rising from ~0 to +0.78**.
- VPoser optimizes a **32-D learned pose space instead of 69 raw joint rotations**. Since that space only contains humanly-possible poses, fits can't drift into impossible configurations, and jitter drops.
- The optimizer runs staged. **Rigid position/orientation first, then full pose**. Solving both together from a rough start lets rigid motion absorb error pose should resolve; staging prevents that and stabilizes convergence.
- Body shape (the SMPL betas) drifted to unrealistic values (± 12.8) once the fit went purely 2D-driven—**raising the shape-prior weight from 0.01 to 5.0** kept body shape physically plausible.

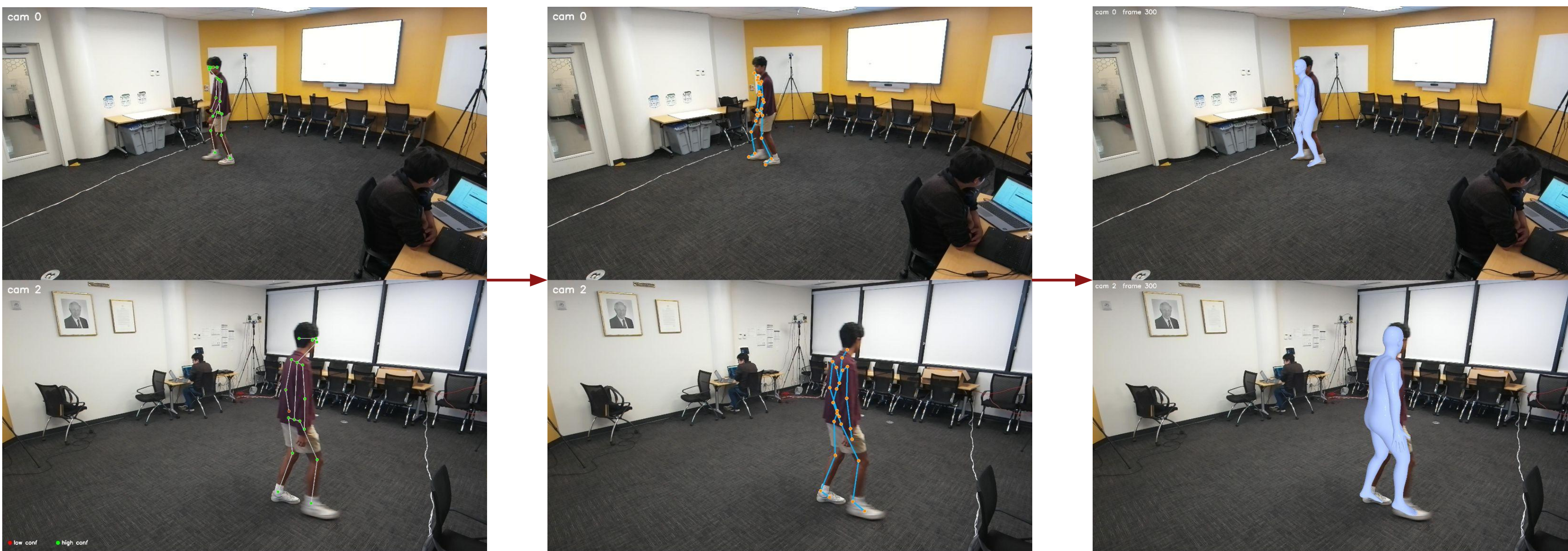


Fig. 4. Ground-truth fitting pipeline, left to right: 2D keypoint detection (BODY25 skeleton, per-frame) → multi-view SMPL+VPoser fit → final rendered mesh reprojected onto the raw frame. Top row: camera 0; bottom row: camera 2 — 2 of the 4 synchronized camera views are shown here for space; the actual fit uses all four.

3 · Results & Next Steps

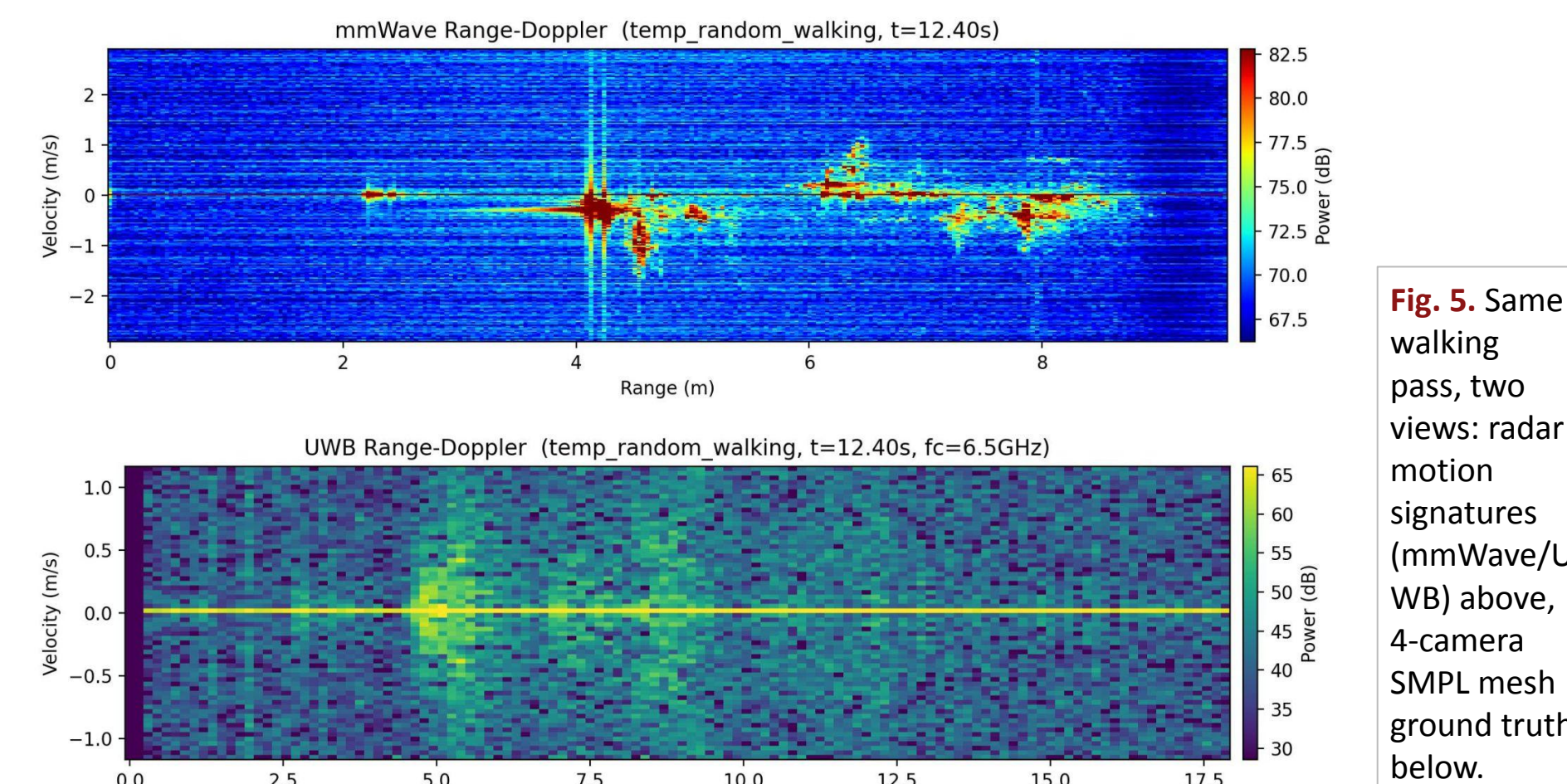


Fig. 5. Same walking pass, two views: radar motion signatures (mmWave/UWB) above, 4-camera SMPL mesh ground truth below.



Fit-consistency across seven sequences

Sequence	Frames	Mean px	cam0	cam1	cam2	cam3
random_walking	602	14.1	17.4	13.3	16.1	9.9
walking_speeds	902	46.5*	20.8	12.1	22.9	136.8*
freezing_gait	601	9.2	9.3	10.0	9.3	8.2
shuffling_gait	601	13.8	16.6	9.4	17.5	11.8
tandem_balance	602	9.0	11.1	10.1	6.6	8.3
turning_90	601	13.8	15.1	9.4	15.7	15.2
corner_reflector	599	21.4	37.8*	19.7	9.0	18.3

Framing note:

fit self-consistency vs. the same 2D detections used to fit it—not validated ground-truth accuracy (no marker-based mocap exists in this project).

Results & next steps

- Achieved: a **fully synchronized capture system** recording four cameras and two radars into one combined, time-aligned rosbag every session.
- Achieved: a validated multi-view SMPL and VPoser fitting pipeline that produces a self-consistent 3D ground-truth mesh directly from the camera views, with **no external motion-capture hardware required**.
- Next: Improve capture and post-processing robustness with a UI that monitors sync health and detection quality **across all 4 cameras in real time**.
- Next: scale data collection to 20 subjects and begin training a radar-only model that **predicts gait activity directly from motion signatures**.

References & acknowledgments

This is the foundational paper our approach builds on for radar-based motion sensing. Choi, Hor, Yang, Arbabian. "MVDoppler-Pose: Multi-Modal, Multi-View mmWave Sensing for Long-Distance, Self-Occluded Human Walking Pose Estimation." CVPR 2025. With thanks to Prof. Amin Arbabian, mentor Shubo Yang, labmate Edison Altamirano, and the Stanford EE REU program.